



Computer Science & IT



Data Structure & programming

Tree

Lecture No. 01



By- Abhishek Sir

Recap of Previous Lecture



Topic

Circular Linked List

Topic

Traversal Add begin Addend

Topic

Stack using Linked List

Topic

Linear Queue



Addend

Topic

$\text{Enqueue} = \text{dequeue} = O(1)$

↑
front

↑
rear

Topics to be Covered



Topic

Tree

Topic

Terminology

Topic

General tree, forest

Topic

Complete k -ary tree

Topic

Theorem

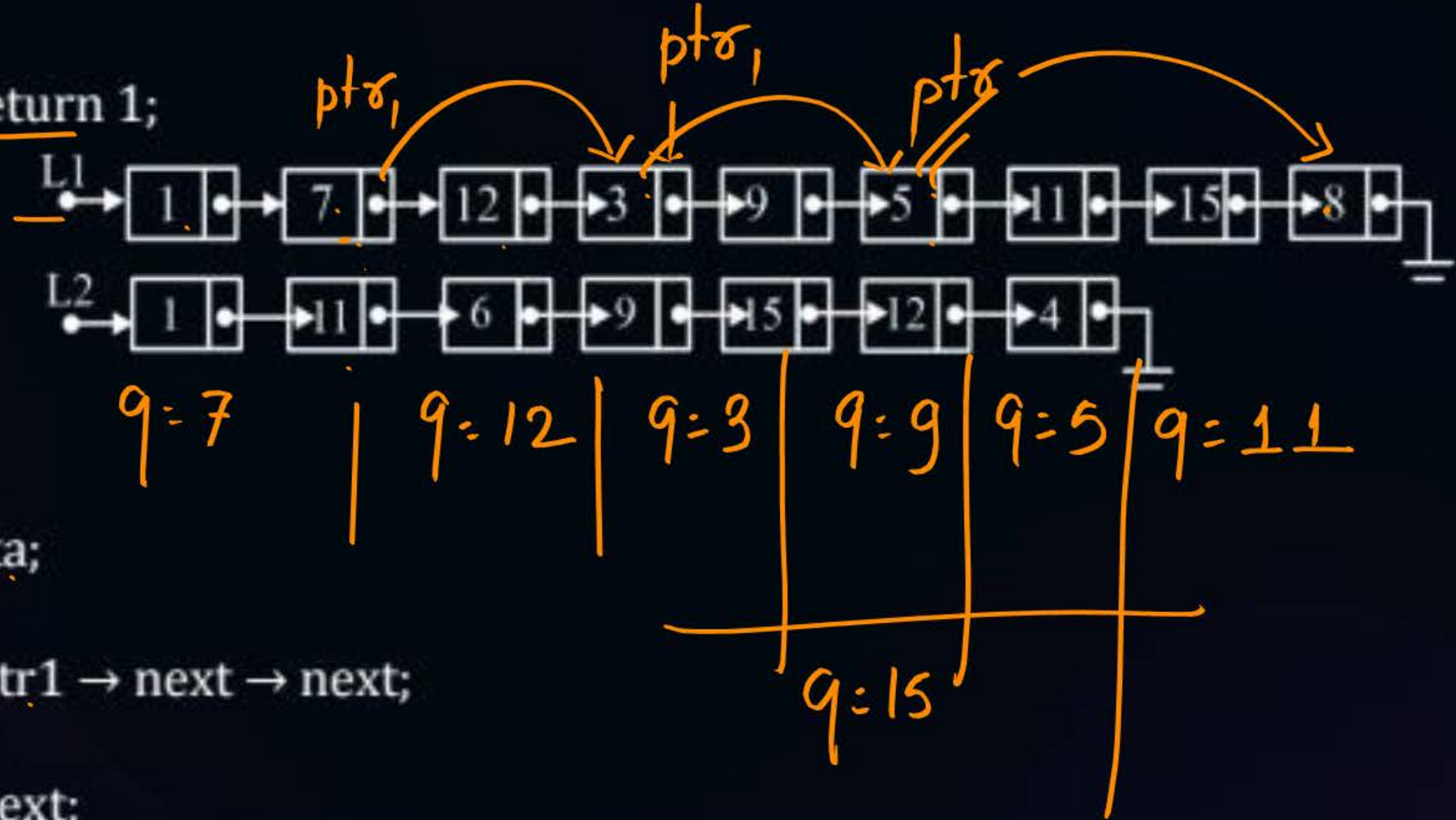
Consider the following ANSI C program:

```
#include <stdio.h>
#include <stdlib.h>
struct Node{
    int value;
    struct Node *next;};
```

Consider the following C-prog segment that modifies the list L1. The number of nodes that will be there in L1 after execution of the code sequence is 5.

```
int find (int query, LIST *list) {
    while (list != NULL) {
        if (list → data == query) return 1;
        list = list → next;
    }
    return 0;
}
```

```
int main (){
    List ptr1 = L1; ptr2 = L2;
    while (ptr1 → next != NULL) {
        query = ptr1 → next → data;
        if ((find(query, L2))
            ptr1 → next = ptr1 → next → next;
        else
            ptr1 = ptr1 → next;
    }
    return 0;
}
```

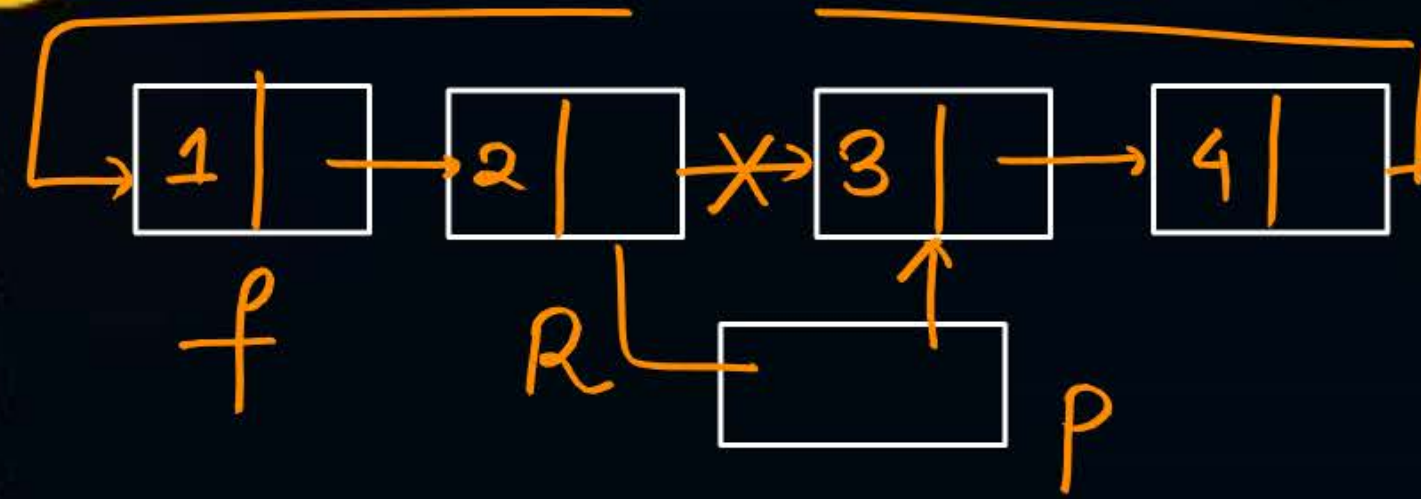




Question

A circular queue has been implemented using a single linked list where each node consists of a value and a single pointer pointing to the next node. We maintain exactly two external pointers **FRONT** and **REAR** pointing to the front node and the rear node of the queue, respectively. Which of the following statements is/are **CORRECT** for such a circular queue, so that insertion and deletion operation can be performed in $O(1)$ time ?

- I. Next pointer of front node points to the rear node.
 - II. Next pointer of rear node points to the front node.
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II



|| Create New Node

$p \rightarrow next = R \rightarrow next$

$R \rightarrow next = p;$

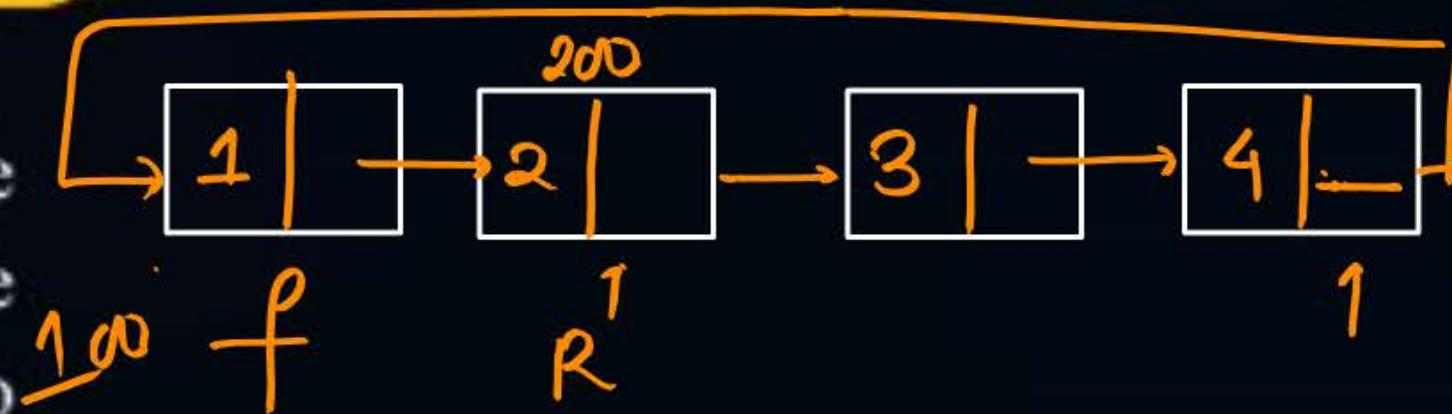


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- II. Next pointer of rear node points to the front node.

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- (B) II only
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- (D) Neither I nor II



for delete Last node (4)

next pointer to update

→ traversal Need $\Rightarrow O(n)$



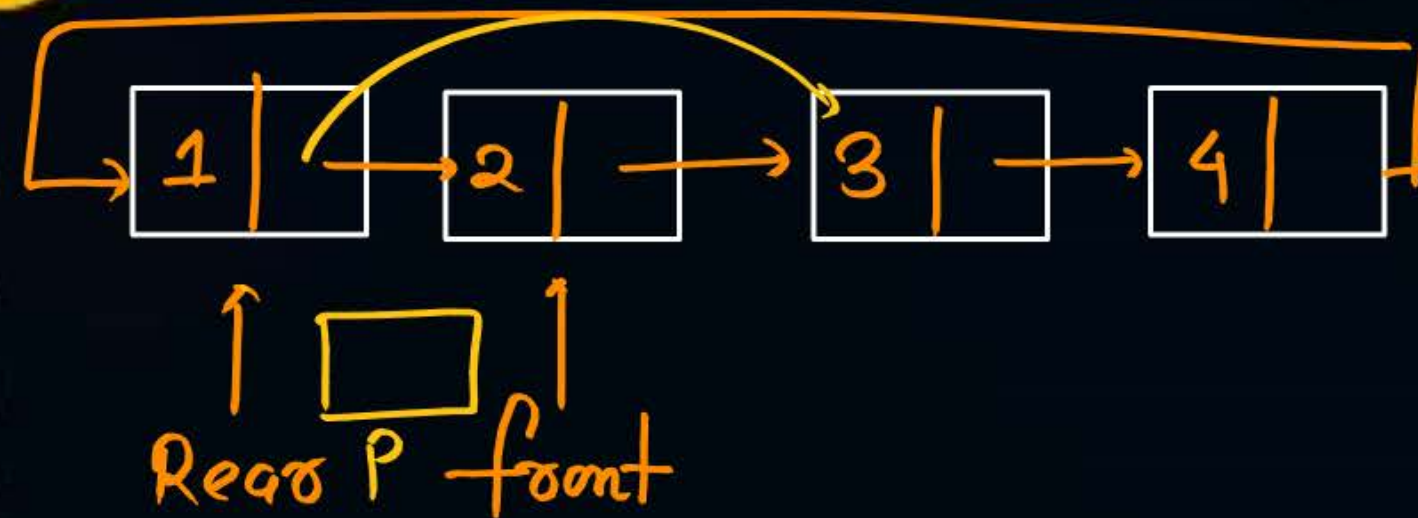
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Insert : Create New Node

$p \rightarrow \text{next} = r \rightarrow \text{next}$
 $r \rightarrow \text{next} = p$ } $O(1)$

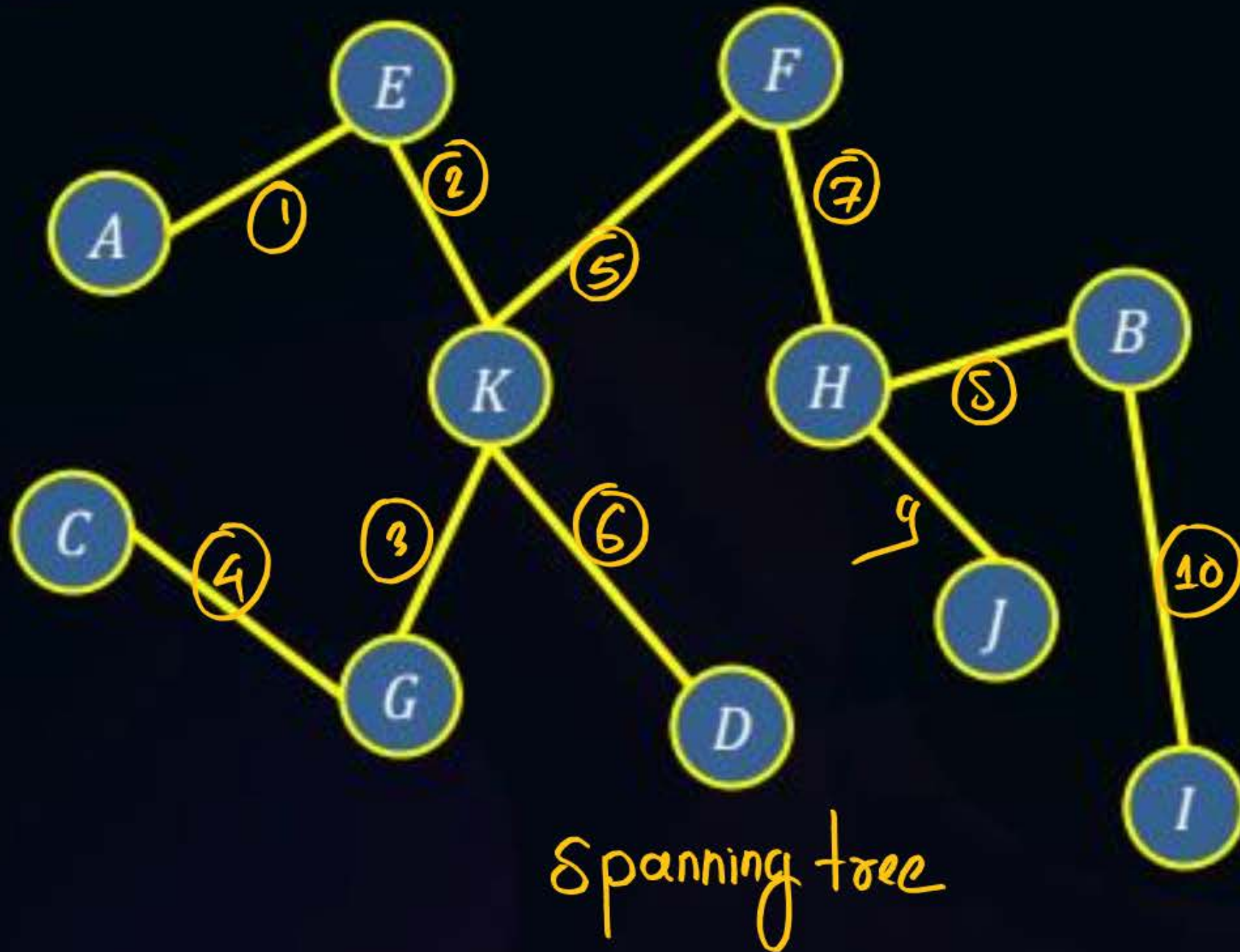
Delete : $\text{Rear} \rightarrow \text{next} = \text{front} \rightarrow \text{next}$
 $\text{free}(\text{front})$
 $\text{update}(\text{front})$

Topic : Tree



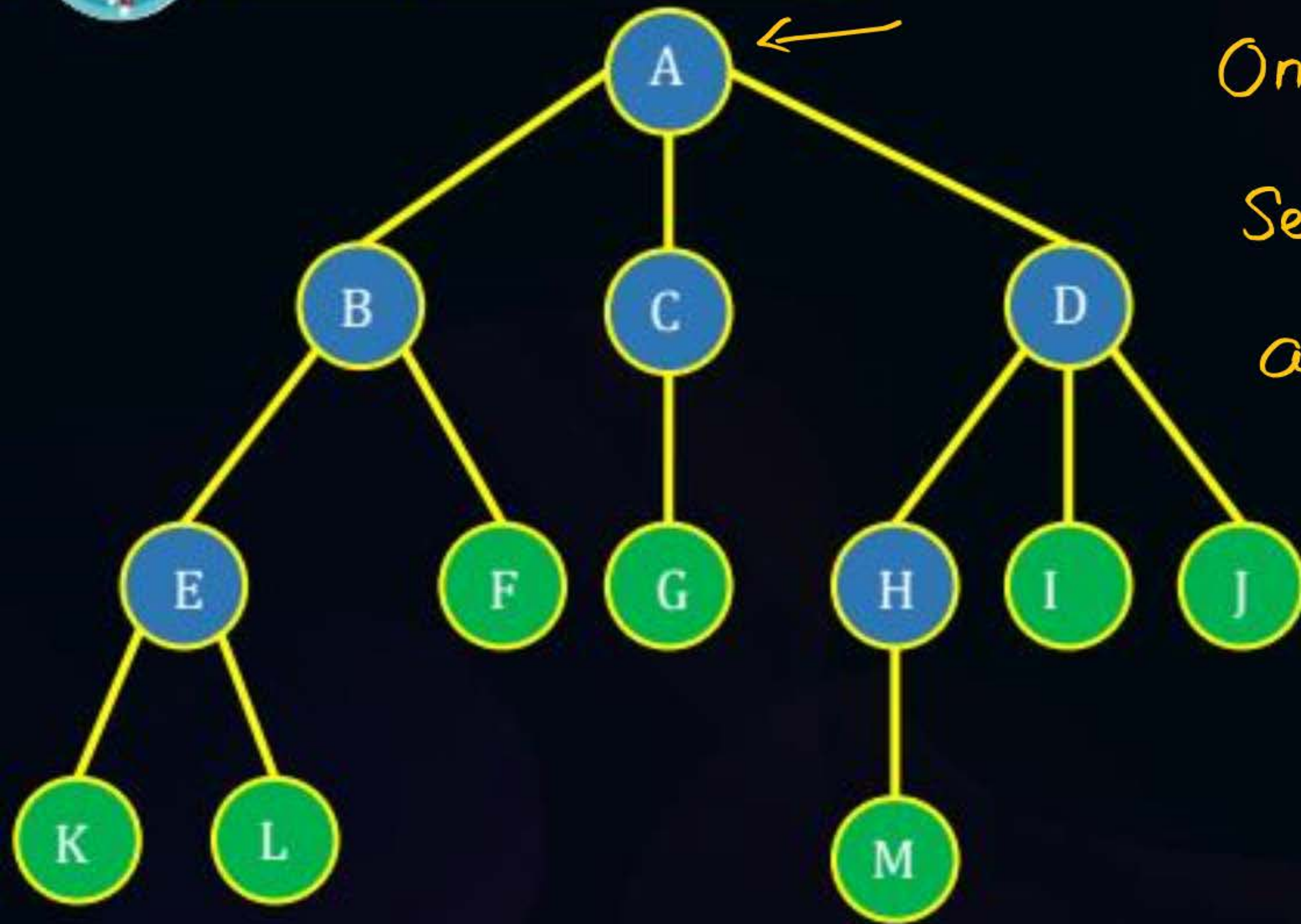
Tree is a special Graph $G(V, E)$

1. Tree is connected graph
2. Tree is acyclic graph
3. A tree with n nodes contains $n-1$ edges.





Topic : Tree



One distinguished node
Selected as Root of tree
and rest of node arranged in
parent child Model.

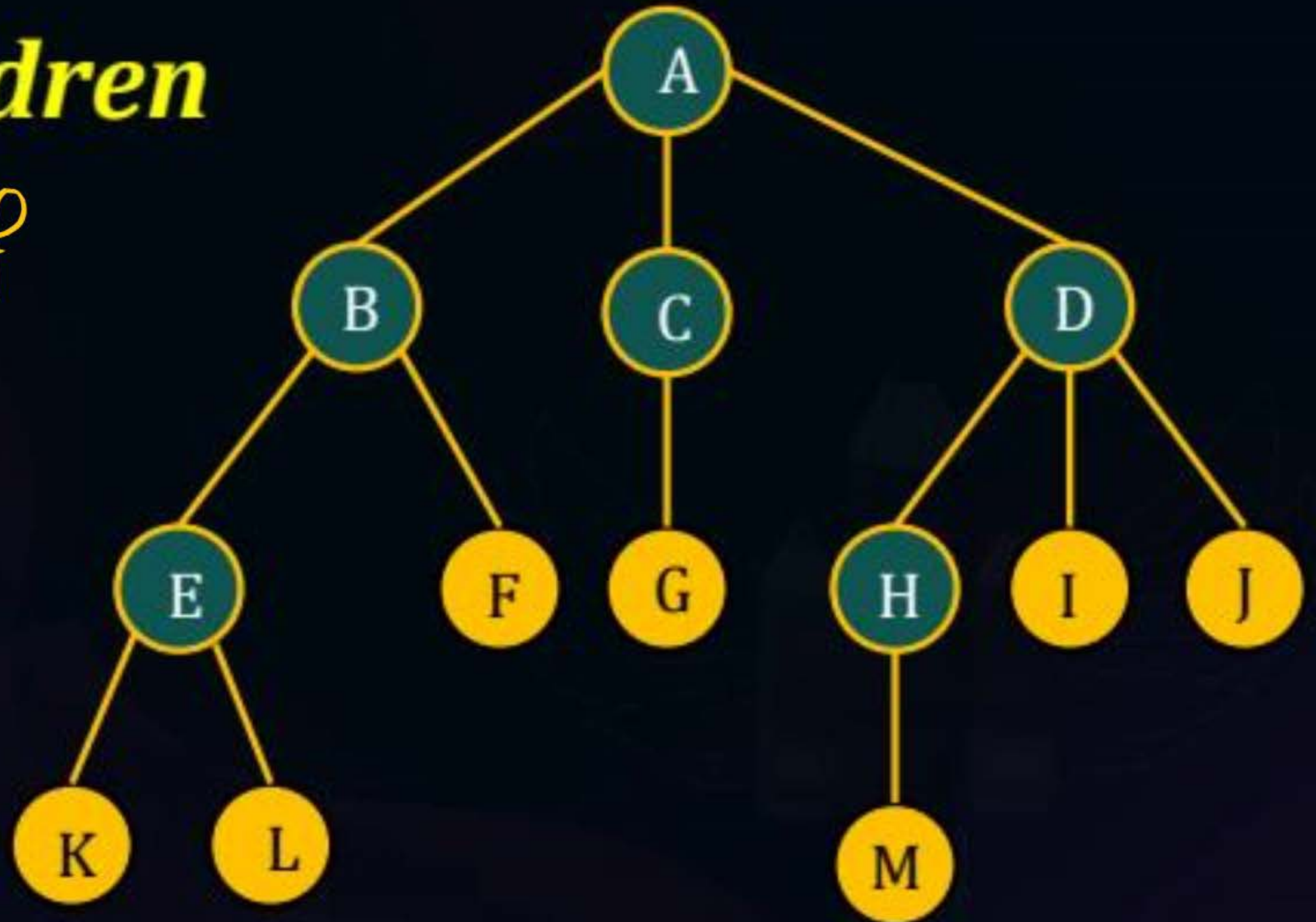


- Parent & children**

D is parent of

H, I, J

*B, C, D are
children of A*

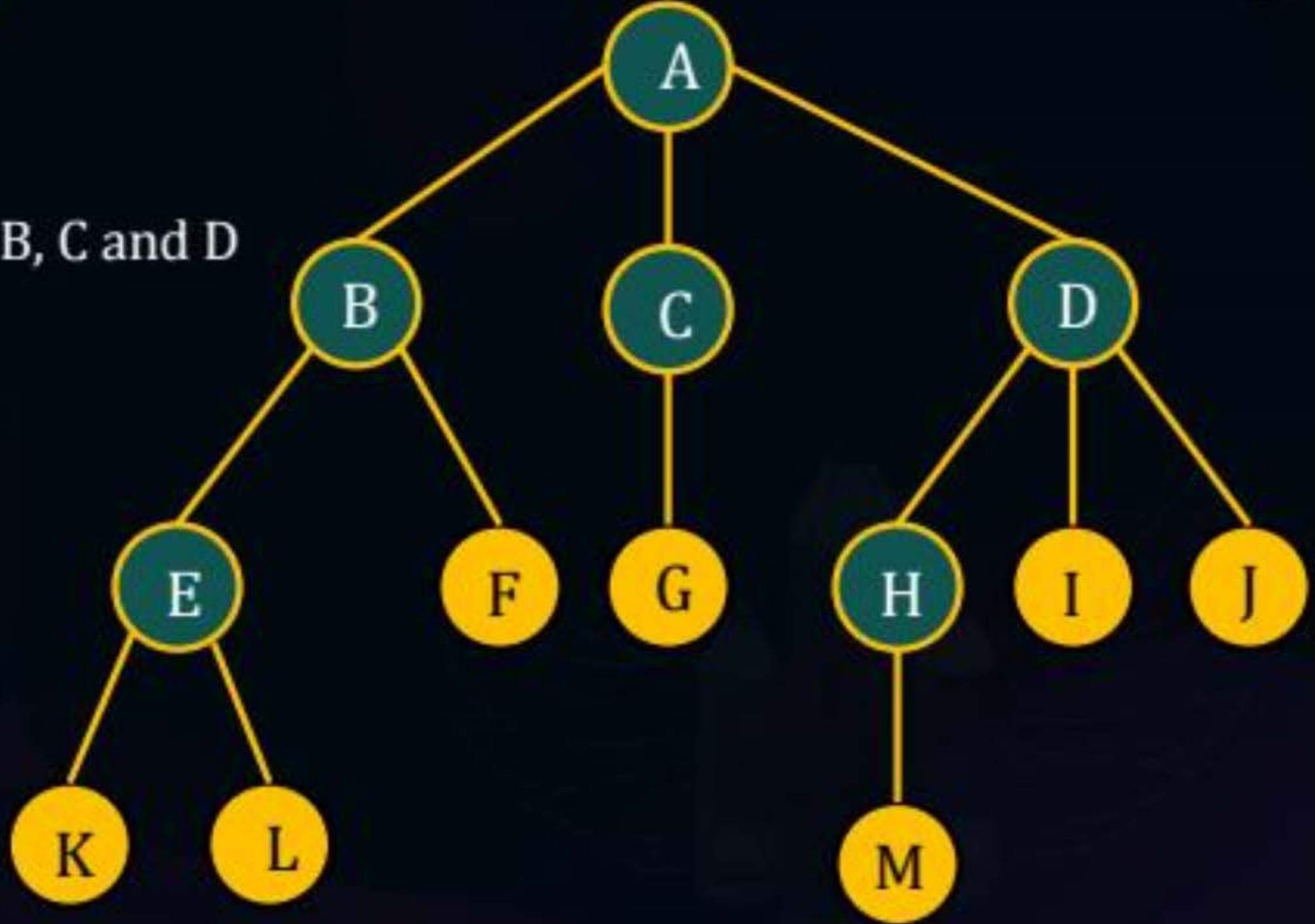




Topic : Tree



- A is the parent of its children B, C and D





Topic : Tree



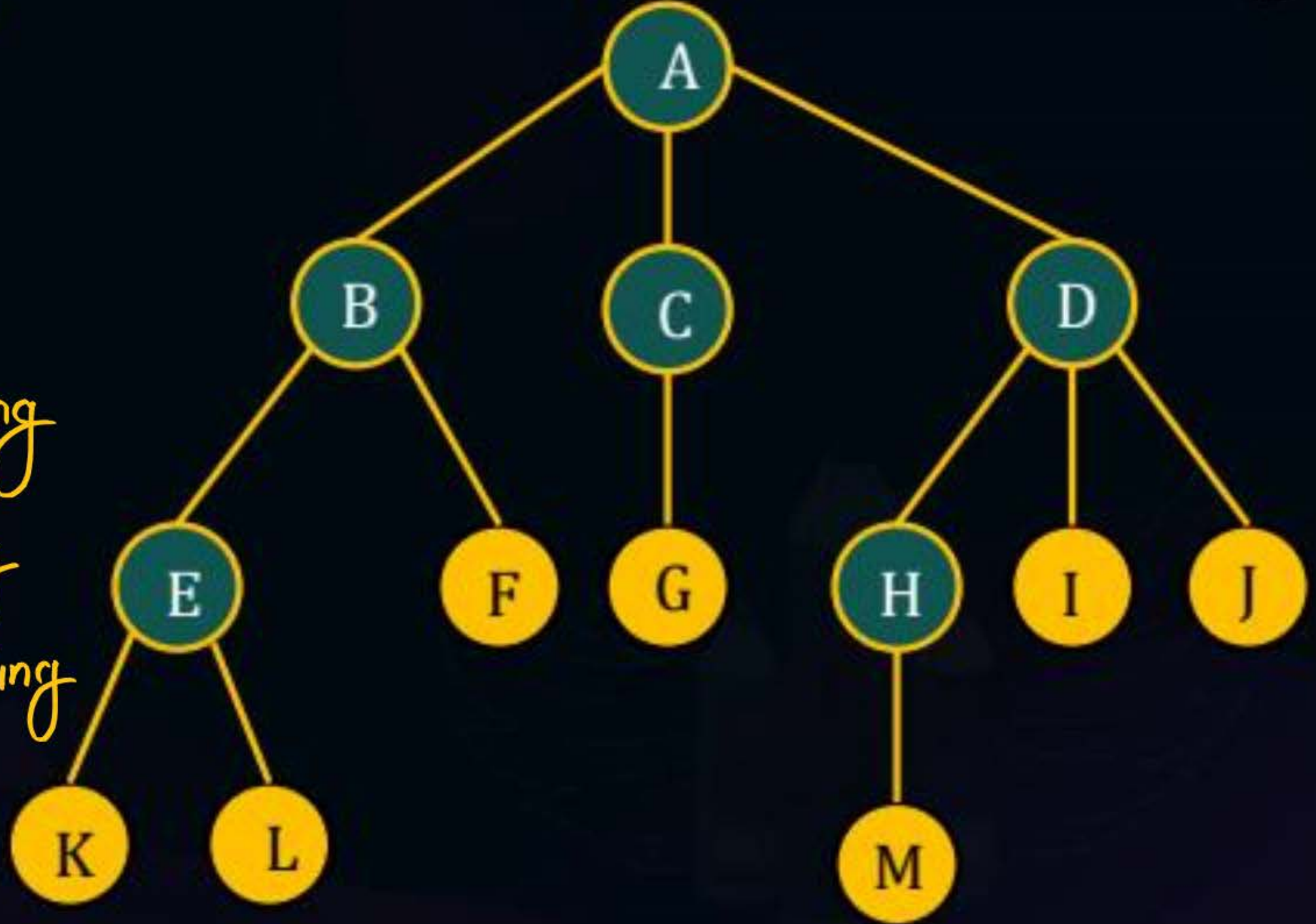
- Siblings.***

Children of same parent.

H, I, J are sibling

K, L are sibling

F, G are not sibling



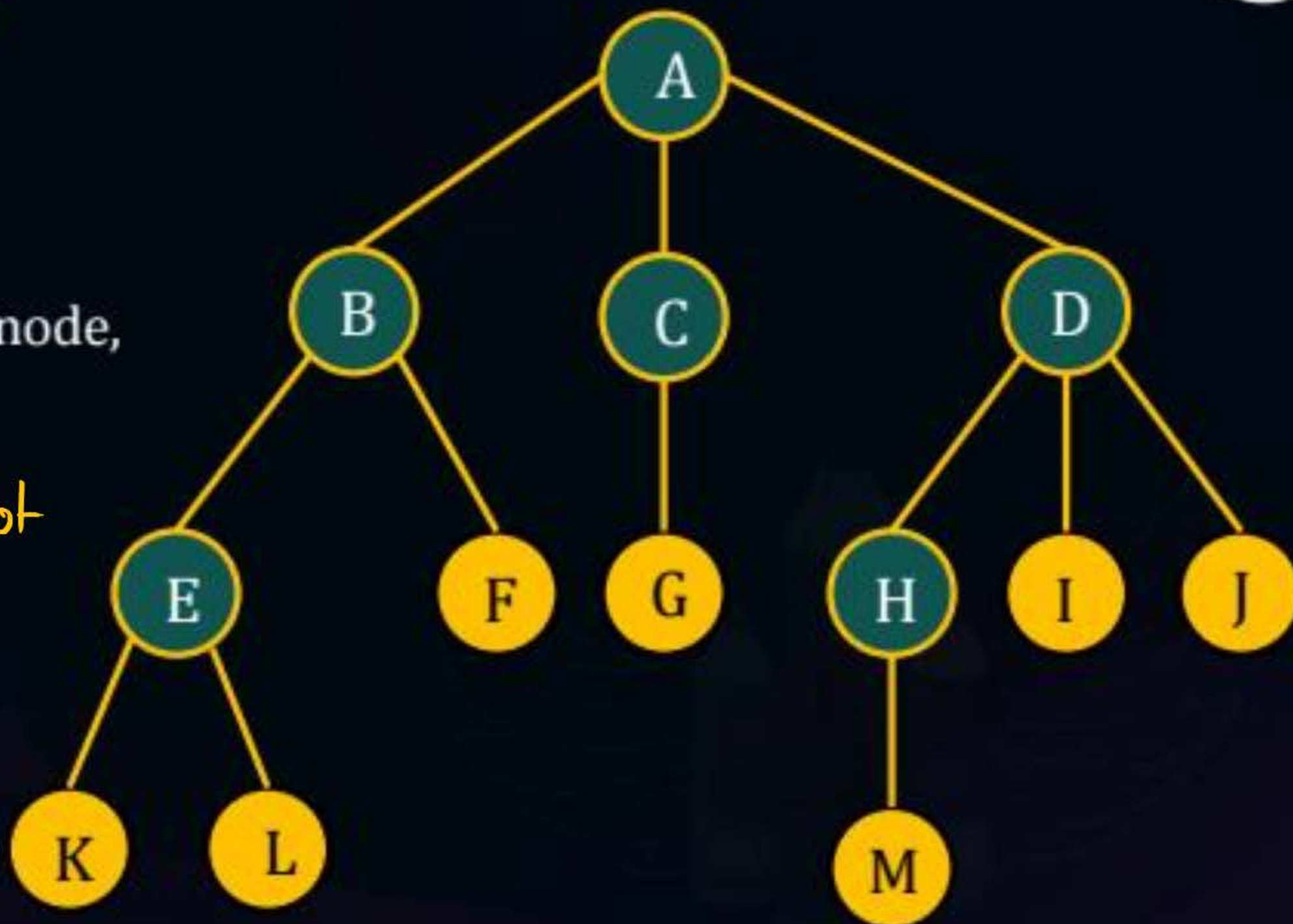


Topic : Tree



- **Root** = distinguished node,

Root node does not
have any parent

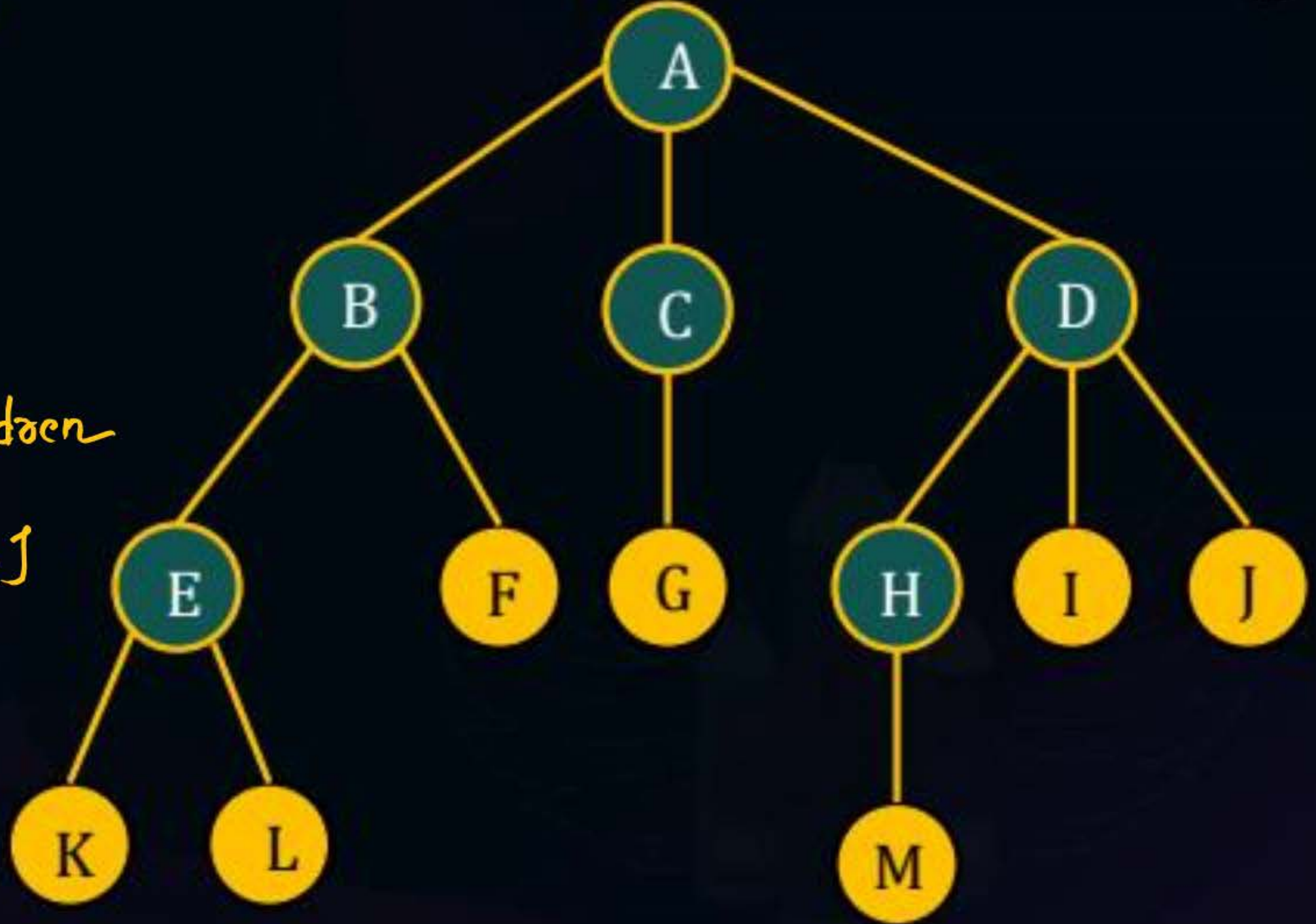




• *Leaf Node:*

Node with No children

K, L, F, G, M, I, J
are Leaves





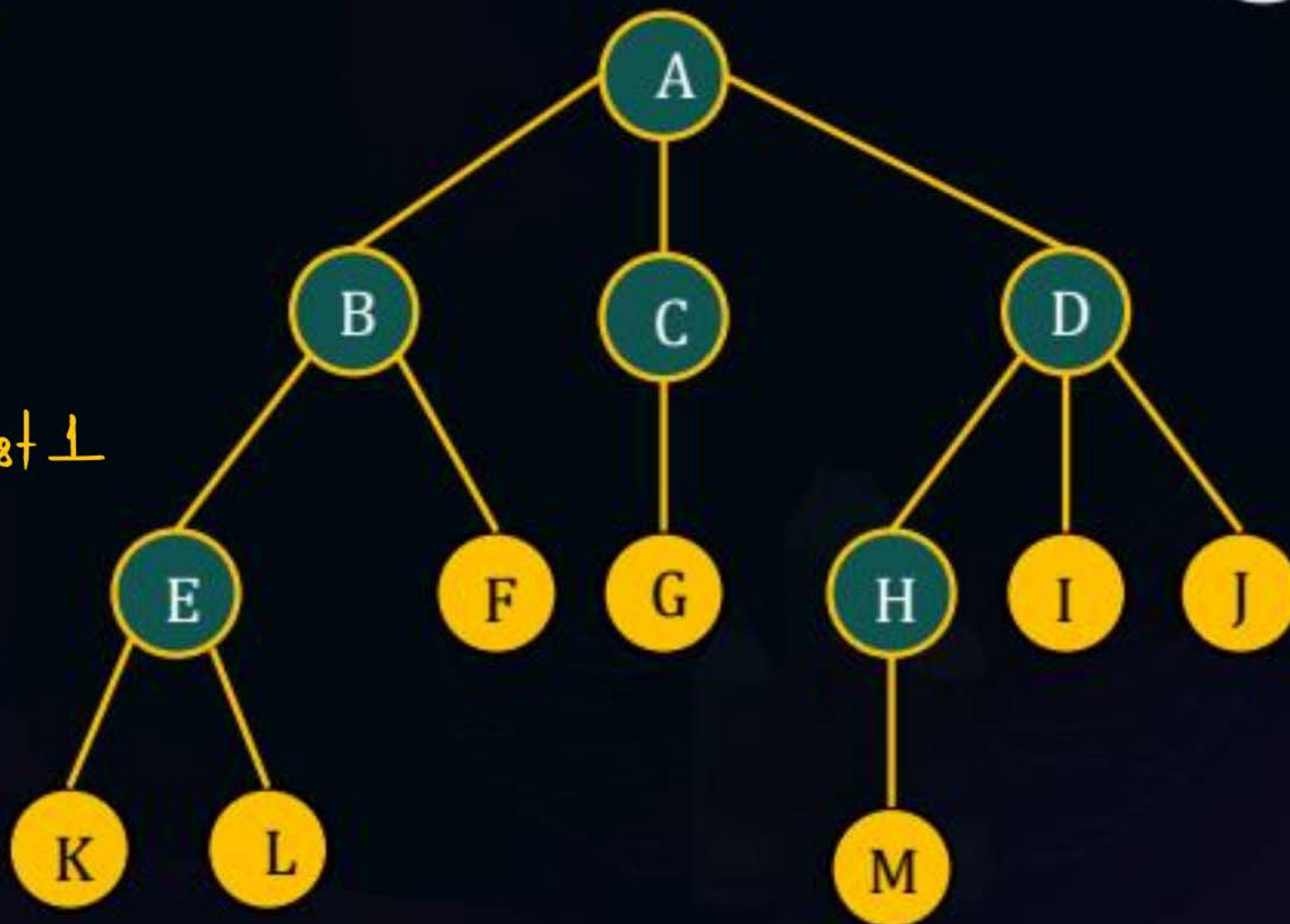
Topic : Tree

Internal Node:

A node with atleast 1

Children

A, B, C, D, E, H



Total No. of Nodes = # Leaves + # Internal Node

Total No. of edges = # of Leaves + # of Internal Node - 1



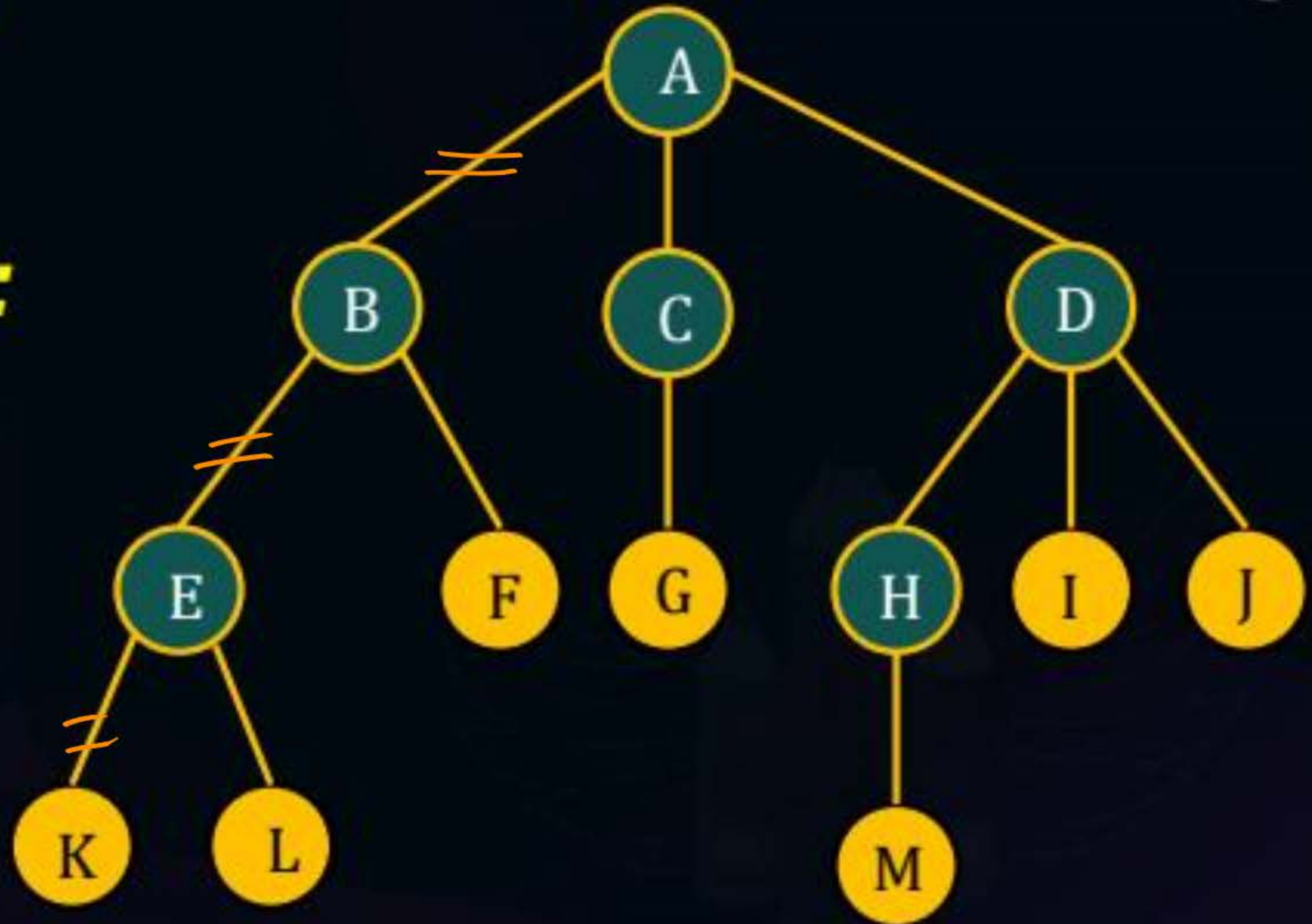
Topic : Tree

• **Height of tree:**

Maximum No. of edges in

Root to Leaf path

height is 3



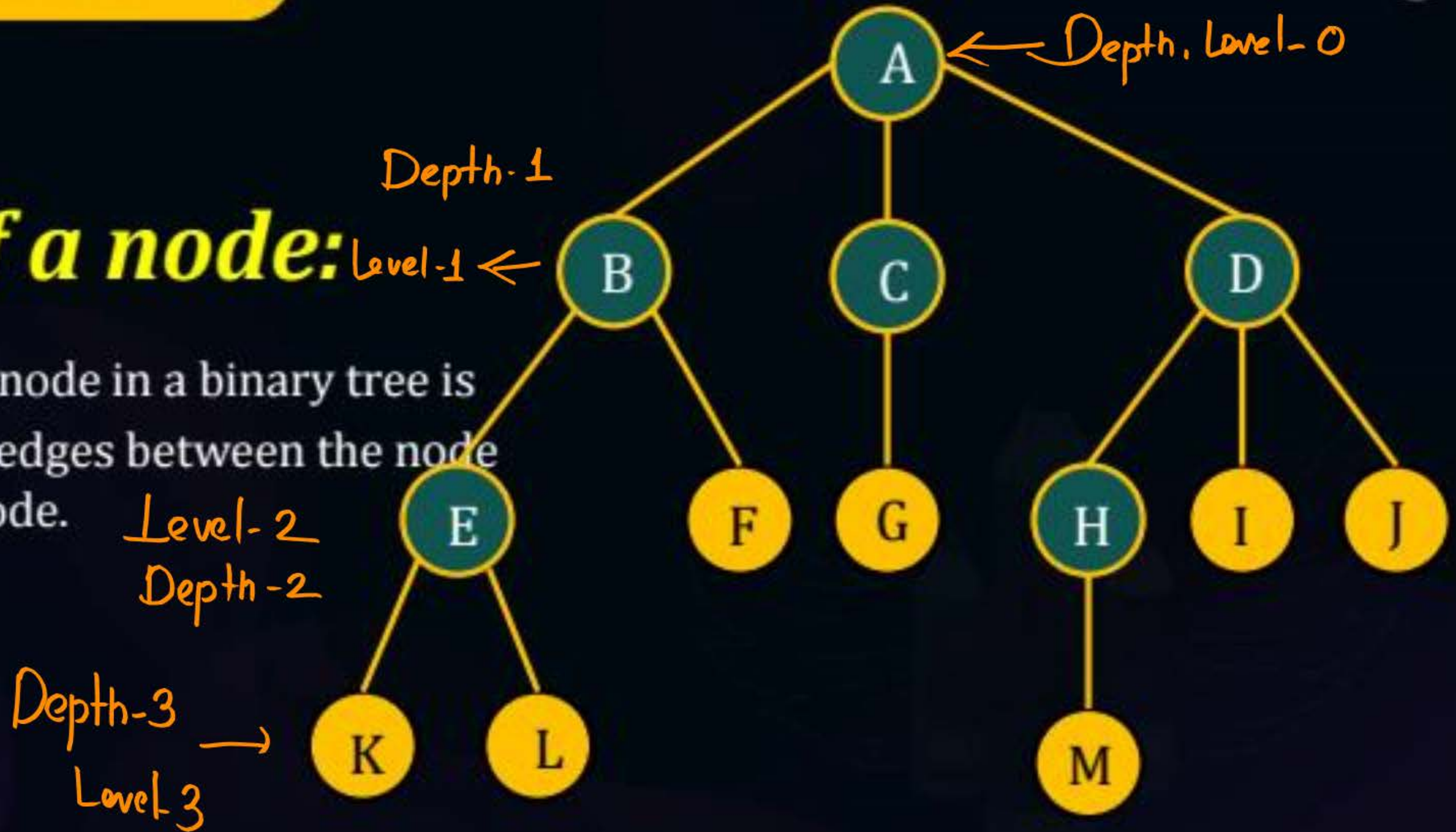


Topic : Tree



Depth of a node:

The depth of a node in a binary tree is the number of edges between the node and the root node.





Topic : Tree

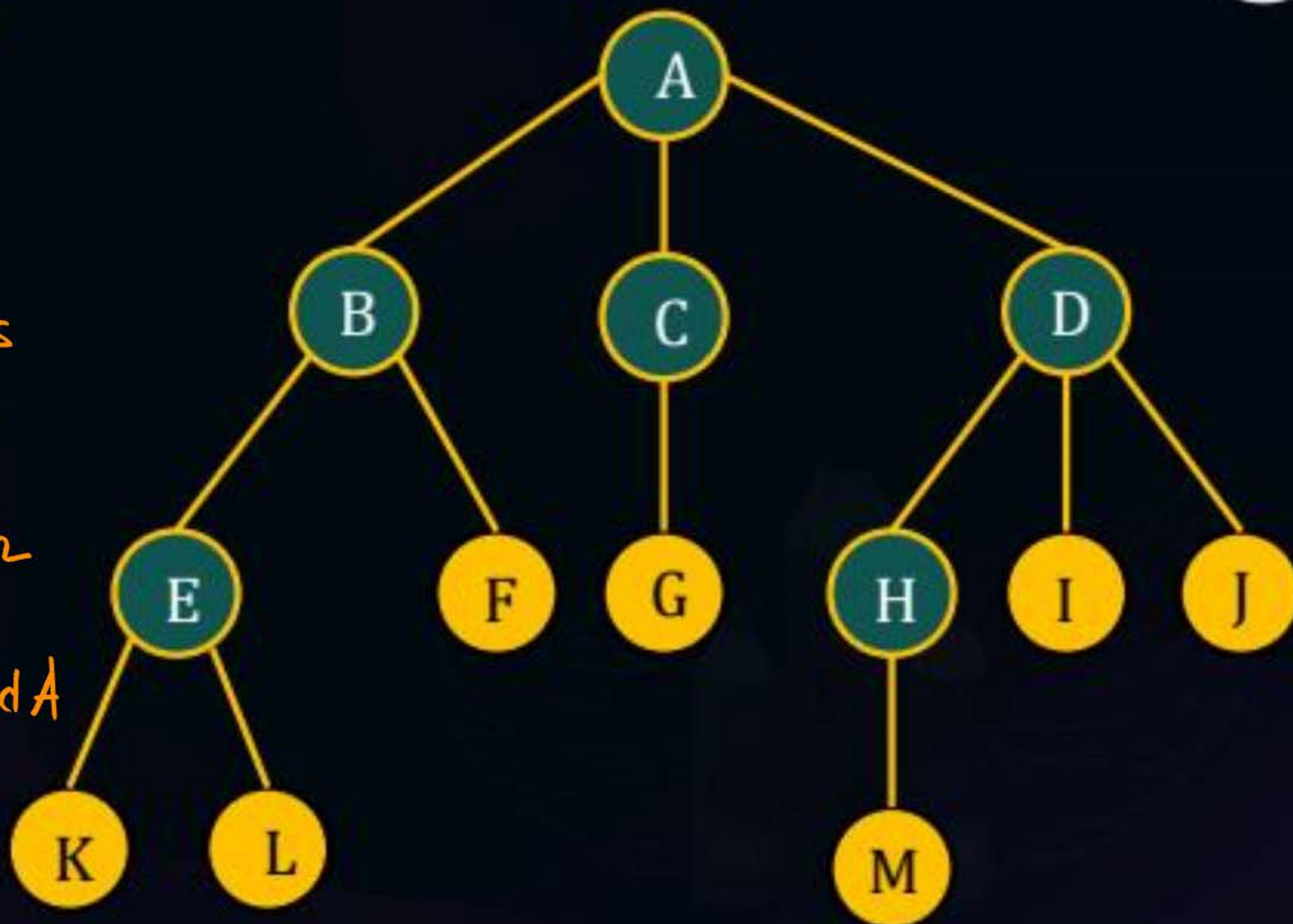


- ancestors***

Ancestor of Node x is
all Node that encounter
in node x to root path

Ancestor F is B and A

Ancestor of M is
H D, A

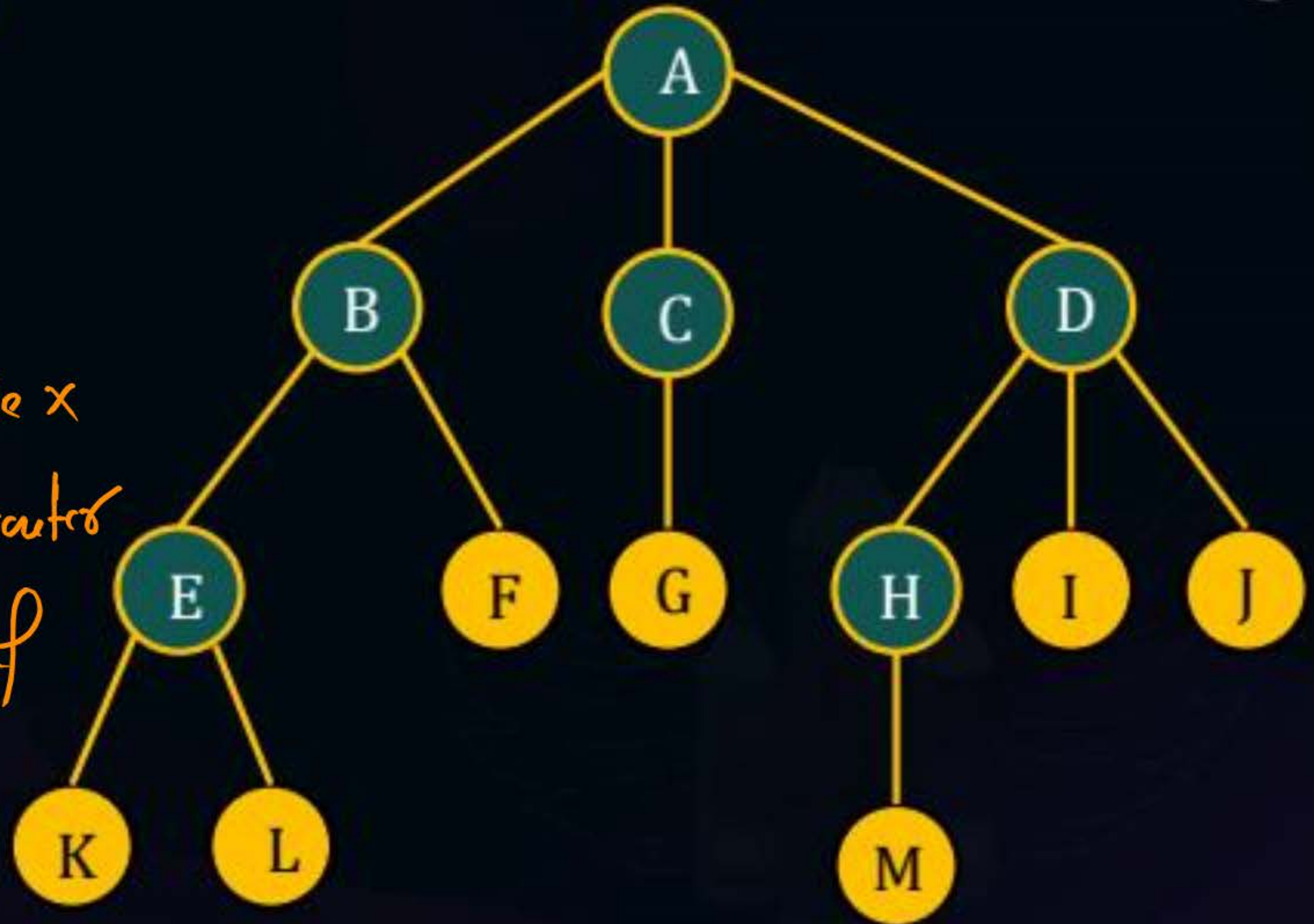




- Descendant***

Descendant of node x
is all node that encounters
in Node x to Leaf
path

E, F, K, L are
descendant of B

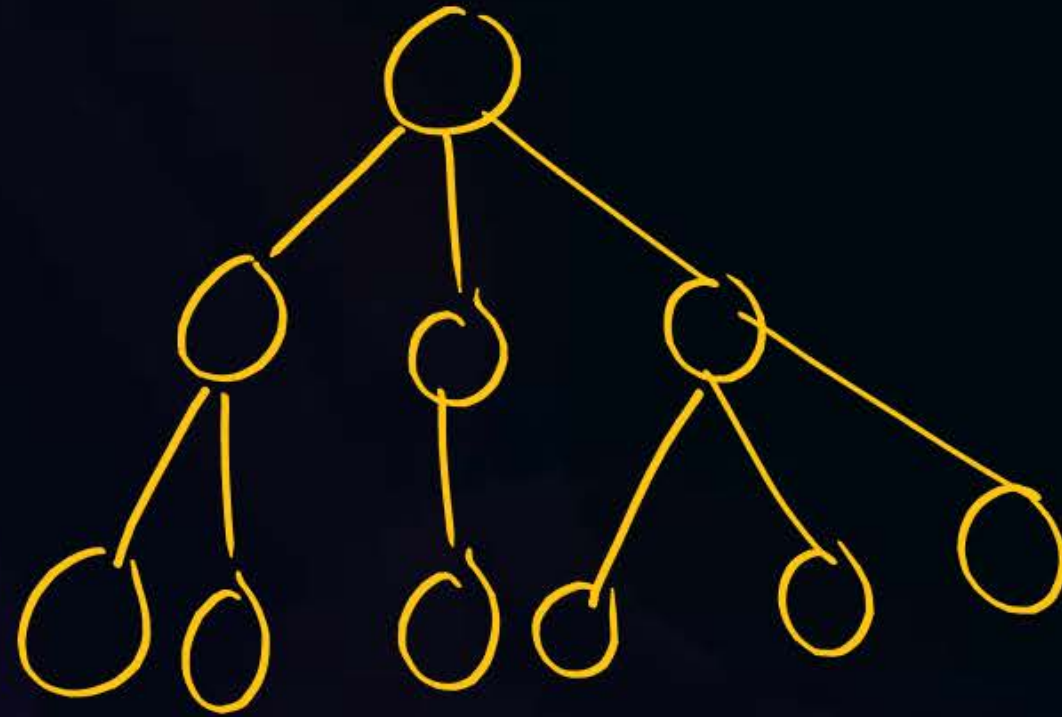




Topic : Tree



General tree: A general tree is a tree in which each node can have 0 or more children.





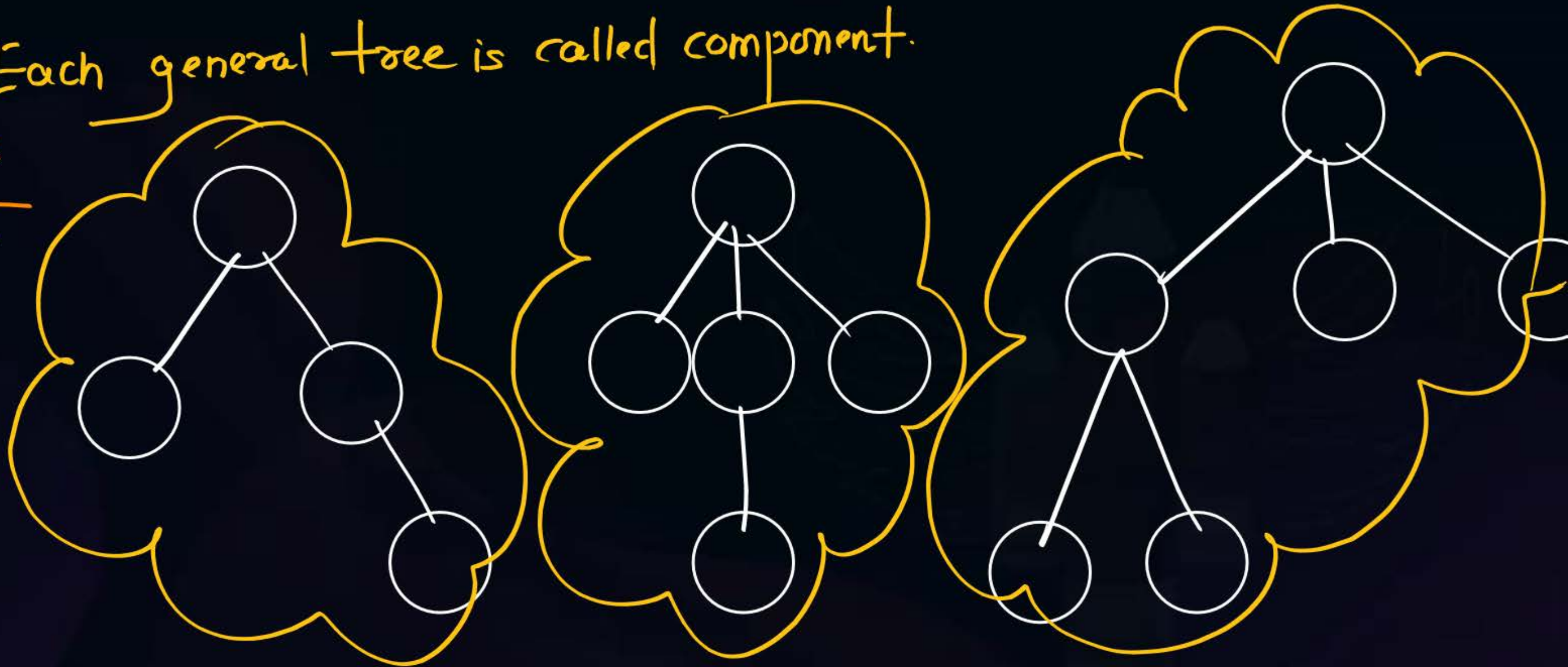
Topic : Tree



forest : forest is collection of one or more general tree

Each general tree is called component.

3 components





Topic : Tree



degree of Node :

Tree is a graph
degree of Node No. of
edges it connected

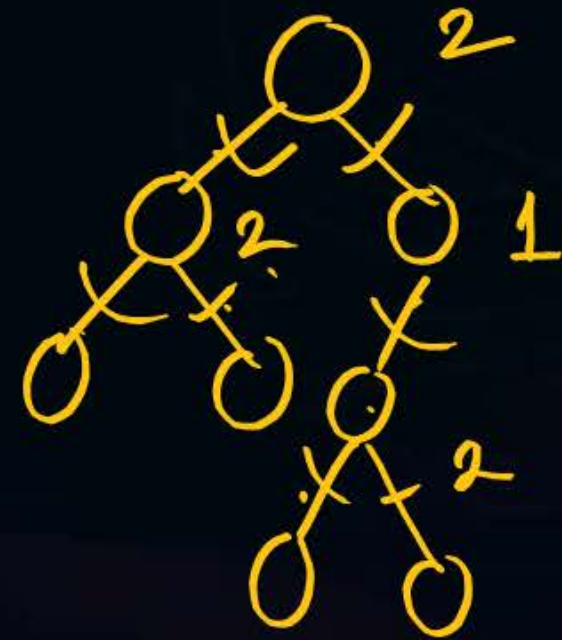
Sum of degree : $2|E|$

2017

degree of Node : Tree

degree of Node is

No. of children a node is having



Sum of degree
= No. of edges



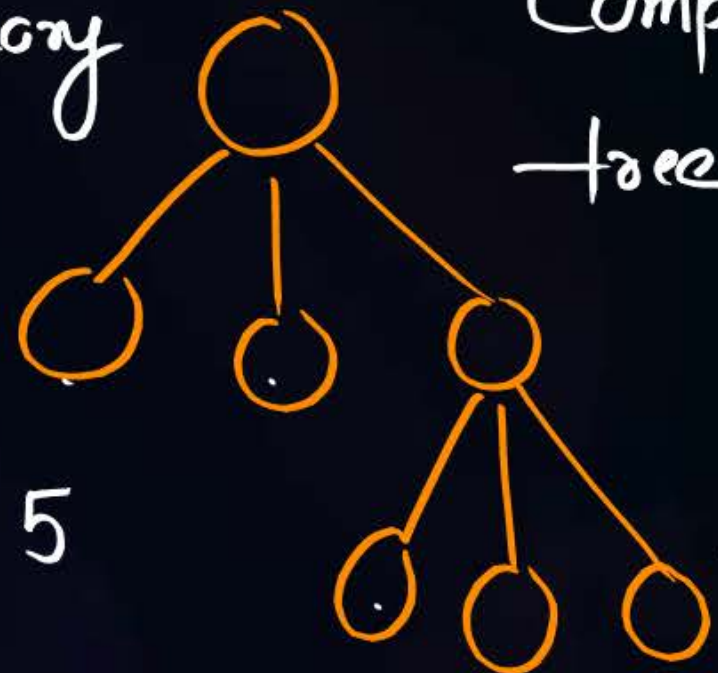
Topic : Tree

Complete k-ary tree

A complete k-ary tree is a tree

in which each node is having 0 or k children.

Complete 3-ary tree



$L = 5$
 $I = 2$

$$L = 2 \times (3 - 1) + 1 = 5$$

Complete 2-ary tree



$L = 4$
 $I = 3$

$$L = 3 \times (2 - 1) + 1 = 4$$

Complete 4-ary tree



$L = 10$
 $I = 3$

$$L = 3 \times (4 - 1) + 1 = \underline{10}$$



Topic : Tree



Theorem: In complete k -ary with L leaves and I

Internal node:

$$L = I(k-1) + 1$$



Topic : Question



#Q A complete n -ary tree is a tree in which each node has 0 or n children. Let I be the number of internal nodes and L be the number of leaves in a complete n -ary tree. If $L = 41$, and $I = 10$, what is the value of n ?

(A) 3

(B) 4

(C) 5

(D) 6

$$L = I(n-1) + 1$$

$$\Rightarrow 41 = 10(n-1) + 1$$

$$\Rightarrow 10(n-1) = 40$$

$$\Rightarrow \begin{matrix} n-1 = 4 \\ \textcircled{n = 5} \end{matrix}$$



Topic : Question



PYQ - 2017

#Q Let T be a tree with 10 vertices. The sum of the degrees of all the vertices in T is _____.

10 vertices

No. of edges = 9

$$\text{Sum of degree} = \underline{2|E|} = 2 \times 9 = 18$$



Topic : Question



#Q Suppose a 6-ary tree is having 16 Internal Nodes. Then the sum of the degree of all vertices is:

$$L = 16 \times 5 + 1$$
$$= 81$$

Graph

$$\text{Total No. of Nodes} = 81 + 16 = 97$$

$$\text{No. of edges} = 96$$

$$\text{Sum of degree} = 2 \times 96 = 192$$



Topic : Question



The number of leaf nodes in a rooted tree of n nodes, with each node having 0 or 3 children is:

Complete 3-ary

(A) $n/2$

(B) $(n-1)/3$

(C) $(n-1)/2$

(D) $(2n+1)/3$

$$L + I = n \quad \text{--- (I)}$$

$$L = I(3-1) + 1$$

$$= 2I + 1 \quad \text{--- (II)}$$

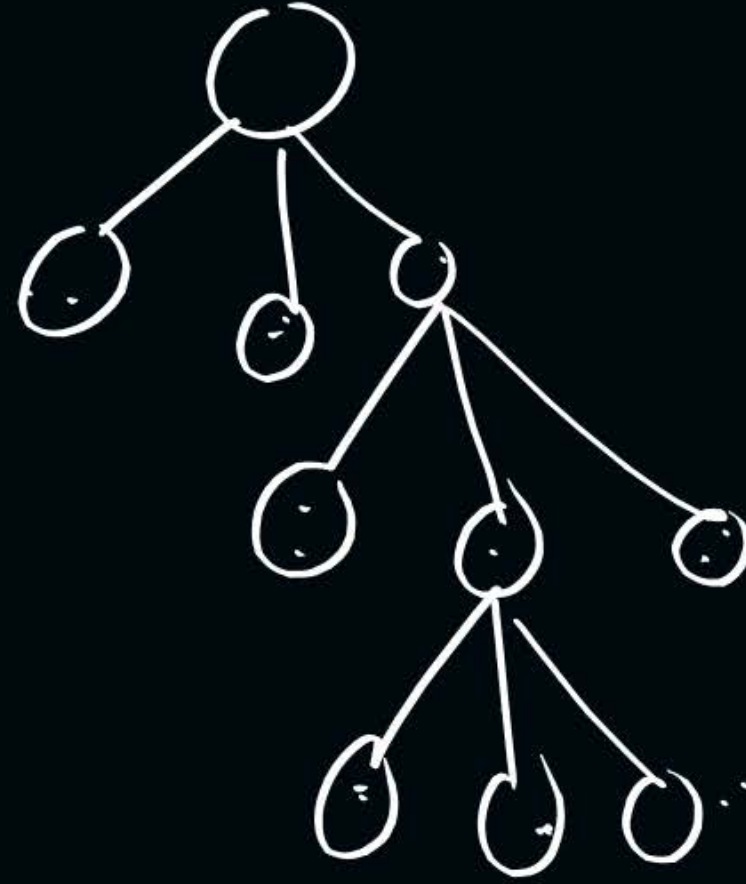
$$I = \frac{L-1}{2}$$

$$L + \frac{L-1}{2} = n$$

$$\frac{2L + L - 1}{2} = n$$

$$3L - 1 = 2n$$

$$\boxed{L = \frac{2n+1}{3}}$$



$$n=10$$

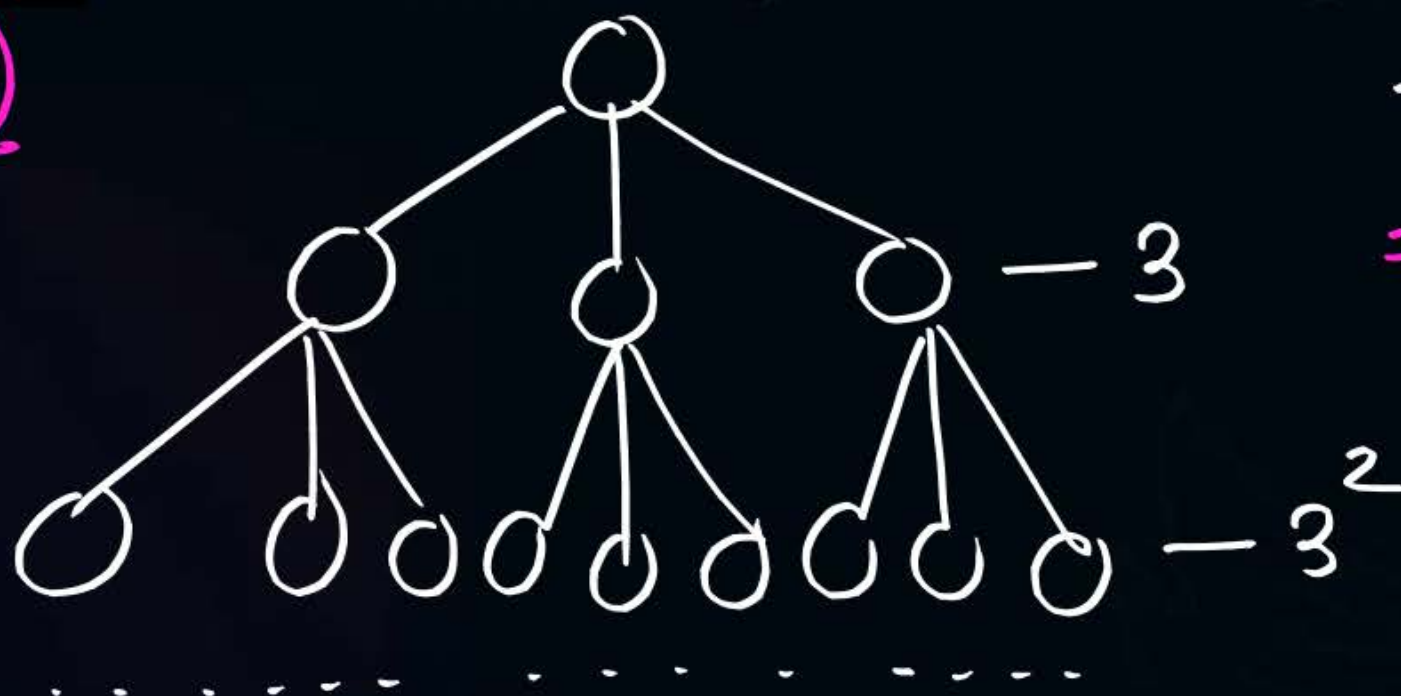
$$L = \frac{2 \times 10 + 1}{3}$$
$$= \underline{7}$$



Topic : Question

#Q Let T be 3 ary tree of height 6 where each node is having 0 or 3 children. The maximum number of nodes possible in T is (The root node is at height 0)?

G.P $\frac{a(r^n - 1)}{r - 1}$
 $r > 1$
 a is first term
 r is ratio
 n No. of terms



total

$$= 3^0 + 3^1 + 3^2 + \dots + 3^6$$

$$= \frac{3^7 - 1}{3 - 1} = \frac{2186}{2}$$

1093

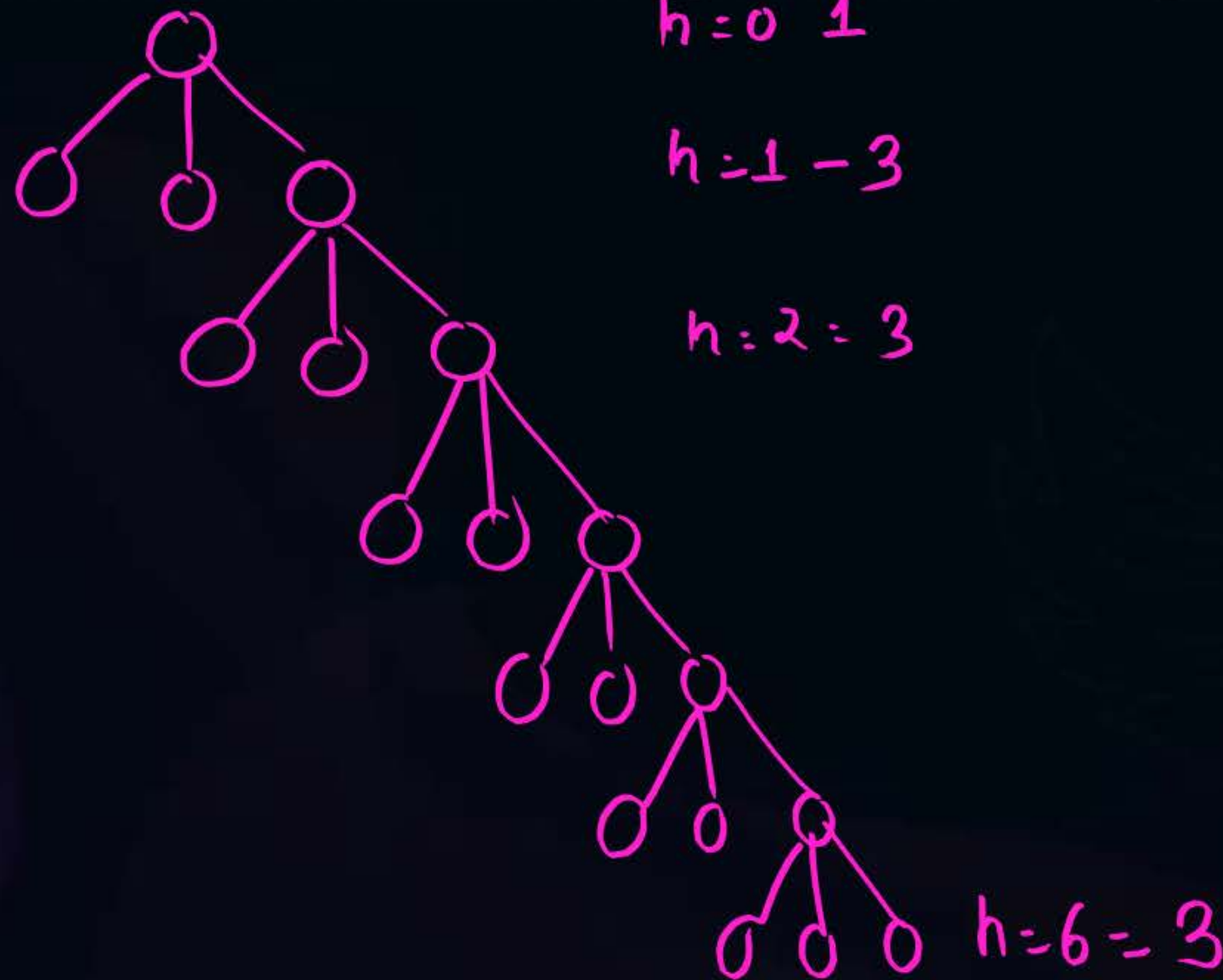
3^h



Topic : Question



#Q Let T be 3 ary tree of height 6 where each node is having 0 or 3 children. The minimum number of nodes possible in T is (The root node is at height 0)?



$$h=0 \quad 1$$

$$h=1 \quad 3$$

$$h=2 \quad 3$$

$$3 \times 6 + 1 = \underline{19}$$

$$\text{Minimum No. Nodes} = \underline{19}$$

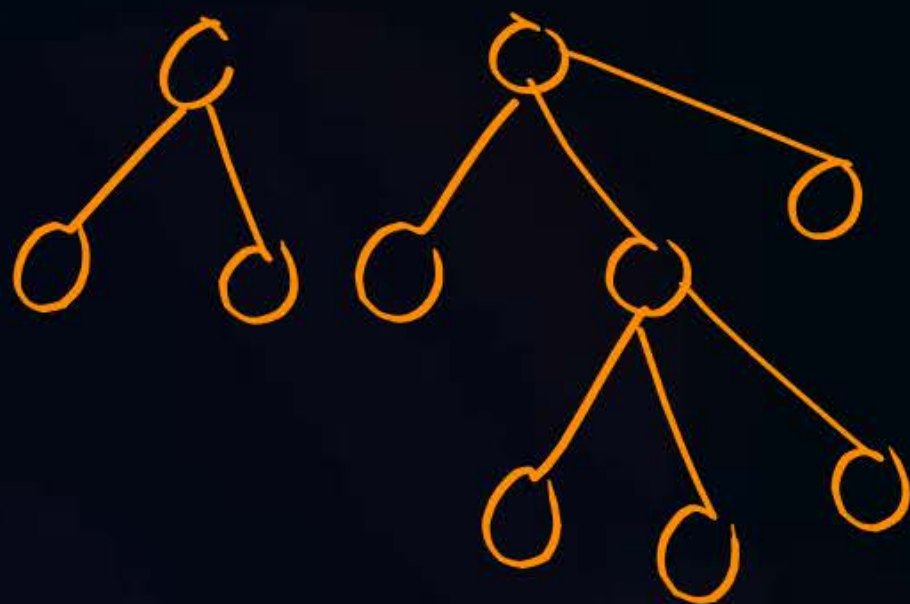


Topic : Question



#Q if a forest consist of total n number of nodes and p component then number of edges in the forest is

- (A) $n-1$
- (B) $n-p$ ✓
- (C) $n+p$
- (D) $n-p+1$



No. of vertex	Component	Edges
n	1	$n-1$
n	2	$n-2$
n	3	$n-3$
n	p	$n-p$ edges

2024 PYQ

Q: The No. of edges present in forest generated by DFS traversal of Graph G

with 100 vertices is 40. No. of connected

component in G is _____

$$\text{No. of edge} = n - p$$

$$40 = 100 - p$$

$$p = 100 - 40 = \textcircled{60}$$



Topic : Question MSQ

#Q which combination of number of leaf node and internal node possible for any value of k , in k -ary tree where each node is having 0 or k children

- (A) $L = 25, I = 4$ ✓ $4 \times 6 + 1 = 25$ - 7 any tree
- (B) $L = 19, I = 2$ $2 \times 9 + 1 = 19$ - 10-ary tree
- (C) $L = 28, I = 3$ $3 \times 7 + 1$ - 8 any tree
- (A) $L = 100, I = 4$ $25 \times 4 + 1$ ✗



Topic : Question MSQ

#Q If a k-ary tree is having height h then the minimum No. of Nodes in the k-ary tree is:

(A) $(k-1)h - 1$

(B) $(k-1)h + 1$

(C) $kh + 1$

(D) $kh - 1$

$$kh + 1$$



Topic : Question



If a k -ary tree is having height h then the maximum No. of Nodes in the k -ary tree is:

- (A) $\frac{k^h - 1}{k - 1}$
- (B) $\frac{k^{h+1} - 1}{k}$
- (C) $\frac{k^{h+1} - 1}{h - 1}$
- ✓ (D) $\frac{k^{h+1} - 1}{k - 1}$

Maxim No. of Nodes

$$\frac{k^{h+1} - 1}{k - 1}$$

THANK - YOU